

## **AI and Generative AI in Education: Critical Understanding and Application**

**1. Define AI and Generative AI with examples. Explain the applications of AI in the education sector.**

**Ans:** AI (Artificial Intelligence): AI is technology that enables computers or machines to think, learn, understand information, and make decisions in ways that are similar to human intelligence. **Ex.** Voice Assistants, Google Maps, Self driving cars, etc.

Generative AI: Generative AI is a type of AI that can create new content such as text, images, music, videos, presentations, or computer code based on the instructions given by a user in the form of prompt. **Ex.** ChatGPT, Google Gemini, Copilot, Khanmingo, Magic tools, Pictory, etc.

- **examples showing how these technologies are applied in educational contexts.**

Artificial Intelligence (AI) and Generative AI support education in many ways. AI enables personalized learning by identifying students' strengths and weaknesses. It also supports **assessment and feedback** by identifying mistakes quickly. Generative AI helps teachers create **lesson plans, worksheets, quizzes, and**

**activities**, making learning more effective, interactive, and student-centred.

## **2. Analyse the implications of the use of Artificial Intelligence in the education sector.**

**Ans:** Artificial Intelligence has significant implications for the education sector, offering both opportunities and challenges. **On the positive side**, AI can support personalized learning by analysing students' performance and providing learning materials according to their individual needs. For example, AI-based learning platforms can identify students who need additional practice and recommend suitable activities. Generative AI can also help teachers prepare lesson plans, worksheets, quizzes, and feedback, improving efficiency and reducing workload.

AI can further improve accessibility through tools such as speech-to-text, text-to-speech, and translation, supporting students with diverse learning needs. Immediate feedback can also help students identify mistakes and improve continuously.

However, these benefits must be balanced against **potential risks**. Overdependence on AI may weaken students' critical thinking, creativity, and independent problem-solving if they simply copy AI-generated answers. AI systems may also

produce inaccurate or biased information, requiring teachers to verify outputs. Research and educational discussions also highlight concerns about student data privacy, cybersecurity, academic dishonesty, and unequal access to technology.

Therefore, AI should be viewed as a supportive educational tool rather than a replacement for teachers. Effective teacher supervision, ethical guidelines, digital literacy, and equitable access are essential for responsible implementation.

**3. Use any generative AI tool for an educational purpose (lesson planning, assessment design, research support, content creation, or learning assistance)**

Ans: I used ChatGPT, a Generative AI tool, for educational purposes, mainly for lesson planning and assessment design in teaching Marathi. I used it to prepare lesson plans, learning objectives, classroom activities, worksheets, and assessments for students. For example, I asked the tool to generate questions based on a Marathi lesson and suggest activities for students with different learning abilities.

My experience was useful and time-saving; however, I understood that AI-generated

responses **cannot always be accepted without verification**. I did not observe any major bias or incorrect information in the final output because **I carefully reviewed the content before using it**. I compared the AI-generated material with the prescribed textbook, syllabus, learning outcomes, and my own teaching experience. I also edited questions and activities to ensure that they were appropriate for the students' age, language level, and classroom context.

Critical evaluation of AI responses is important because AI may **sometimes provide inaccurate, incomplete, biased, or misleading information**. Educators and learners must therefore verify information, use reliable sources, and apply their own judgment. AI should support teaching and learning, but **human supervision and critical thinking remain essential for ensuring accuracy, fairness, and meaningful learning**.

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